A Unified Framework for Generating Diverse and Stealthy Adversarial Patches Against Aerial Object Detection

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***Abstract*—Deep neural network (DNN) has become important in the aerial object detection field. However, due to the vulnerabilities of DNN, adversarial patch is proved to be an efficient method to attack the DNN models, which can maliciously manipulate model predictions, potentially causing critical perception and decision-making errors in safety-sensitive systems. Current adversarial patch methods, which directly adopt pixel-level optimization, can obtain a good attack performance, but the generated patches inevitably introduce abstract content that significantly different from the surrounding environment, creating easily identified patterns that undermine stealthiness. Moreover, an optimized adversarial pattern is single and fixed after training, making it easy to defend against. In this work, we propose a unified adversarial attack framework, named diverse and stealthy adversarial patch (DSAP). The DSAP can generate an arbitrary number of patches with different patterns, which are visually integrated with the surrounding environment and difficult to recognize. By placing the patches on the ground, the detector will identify the patches as nonexisting target objects. We train a generator to implicitly encode adversarial content, which can directly map distinct latent vectors to patches with different adversarial patterns, making them difficult to defend against. In addition, we construct a source-domain image set based on the scene background and use a discriminator to constrain the similarity between generated patches and background environment, ensuring stealthiness while keeping attack performance. The extensive experimental results demonstrate the superiority of our patch in terms of stealthiness and defense difficulty compared to ordinary adversarial patches, while the physical-domain experiments indicate that our attack can transfer from digital to physical domain, posing a threat in the real world.**

***Index Terms*—Aerial object detection, generative adversarial network (GAN), physical adversarial attack, remote sensing.**

I. Introduction

DEEP neural networks (DNNs) have achieved a high-quality performance in critical remote sensing applications [1], [2], [3], [4], [5], [6]. Particularly in object detection, methods like UIU-Net [7] and LSKNet [8] have substantially improved detection accuracy. Their contributions enhance the utilization of deep learning technologies in the remote sensing field. Meanwhile, uncrewed aerial vehicles (UAVs) have become one of the important platforms in intelligent remote sensing perception through their submeter spatial resolution, flexible trajectory planning capabilities, and low-altitude operational advantages. The use of DNN models on UAVs for remote sensing data analysis has become a mainstream method, providing decision support for target reconnaissance, disaster emergency response, and ecological monitoring tasks. Despite the great success of DNNs, researches have revealed their vulnerability to carefully crafted adversarial perturbations [9], [10], [11], where malicious adversarial attacks can manipulate deep models' outputs. According to different attack effects, attack methods are broadly categorized into two types, namely, vanishing attack [12], [13], [14] and creating attack [15], [16]. The former can cause the detector to lose detection of specific objects, while the latter causes detection of nonexisting objects, potentially triggering unnecessary automated responses through decision-making errors. Currently, there is limited research on the robustness of DNNs in the field of aerial object detection. These potential research gaps have prompted us to study and develop more diverse adversarial attack methods, understand the mechanism of vulnerability of deep learning models, and provide a basis for improving the robustness of models in the aerial object detection field.

Adversarial attacks targeting remote sensing imagery primarily fall into two categories: adversarial perturbations [17], [18], [19] and adversarial patches [20], [21], [22]. The former generates adversarial examples through slight pixel-level perturbation across entire images, which is hard to deploy in the physical world. In contrast, adversarial patch attacks optimize printable patterns embedded within specific regions of images, which are easier to apply in the real world and attract extensive research attention. Current adversarial patch optimization frameworks predominantly follow a unified pipeline, including initializing random patterns, rendering patches into target images, computing model output through forward propagation, and optimizing patch pixel values via loss constraints. *However, such paradigms often yield abstract and uncontrollable patterns that are easily been perceived by defense systems.* This limitation underscores the need for adversarial patch stealthiness. First, adversarial patches must

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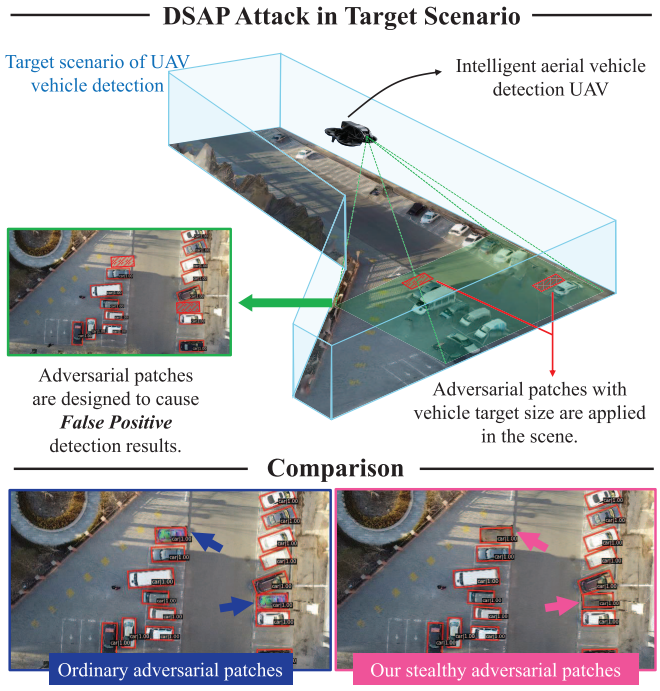


Fig. 1. Illustration of our proposed DSAP attack, with comparison of ordinary adversarial patches and our stealthy adversarial patches. Our method produces patches with diverse patterns that seamlessly integrate into their surrounding environments, achieving superior stealthiness compared to ordinary adversarial patches.

simultaneously achieve attack effectiveness and escape from defense systems. The existing defense algorithms like [23] and [24] are based on images, and thus, patches with visual stealthiness can escape from the perception of defense methods. Second, in human–machine perception-decision systems where critical judgments require expert verification, stealthy attacks can deceive both DNN models and human analysts, inducing dual interferences that amplifies decision-making vulnerabilities.

Considering the stealthiness mechanism of adversarial patches, they can be categorized into semantic-consistent patches and texture-consistent patches. The former achieves stealthiness by aligning adversarial patterns with scene semantics [25], [26], [27], [28]. These methods either utilize pretrained generative models and search for specific latent vectors to generate adversarial patterns or formulate specific semantic-aware loss functions to constrain patch optimization. The former leverages the human visual system's cognitive bias toward contextual plausibility, while the latter simulates the surrounding background textural to ensure continuity in pixel space. *While semantic-consistent patches demonstrate superiority in natural imagery scenarios, this approach fails in aerial images, which usually cover a large area and complex scene elements.* During observation, the human eye relies more on texture continuity rather than local semantic rationality for perception. Moreover, defense methods based on perception models predominantly analyze low-level textural features rather than high-level semantics, making texture-consistent patches achieve a better stealthiness.

We propose a diverse stealthy adversarial attack for aerial object detection systems, as illustrated in Fig. 1. By placing adversarial patches on the ground, they can deceive a UAV aerial image object detector into recognizing the adversarial patches as spoofing target objects. The generated patches exhibit pattern diversity to seamlessly blend with localized background textures across varied positions, enabling a multi-location stealthy patch deployment. Moreover, the patches can transform from the digital domain to physical domain, fooling object detectors in the real world. To achieve these goals, we explicitly incorporate scene-specific contextual information during optimization, enforcing pattern similarity between adversarial patterns and background texture. On the other hand, we utilize a generative model to produce diverse adversarial patches.

In this work, we propose an adversarial patch generation framework based on a generative adversarial network (GAN) that produces diverse adversarial patches through latent vector mapping. We construct a source-domain image set related to the scene background. To achieve stealthiness of adversarial patterns, we constrain them to resemble source-domain images, thereby associating adversarial patterns with scene information. The patches generated in this way can be seamlessly integrated into environmental contexts while maintaining visual consistency with the background environment. Moreover, we add transformations during the training process to bridge the gap between digital and physical domains, enabling the generated patches to be applied in the real world. The main contributions of this work can be summarized as follows.

1) *We are the first to inject background style information into the adversarial patch generation process in aerial image adversarial attack.* By constructing a scenario-related source-domain image set, we explicitly guide the generation of adversarial patches that resemble source-domain images, thereby making our patches achieve a seamless integration into local surrounding environments.
2) *We propose the diverse and stealthy adversarial patch (DSAP) framework to generate diverse adversarial patterns, each resembling the background while remaining deceptive.* Through a single forward propagation, DSAP generates stealthy patches for different locations within the scene, significantly enhancing deployment practicality and flexibility.
3) We conduct *extensive experiments including attack efficacy, stealthiness, diversity, and defense difficulty* to comprehensively demonstrate the effectiveness. Digital experiments along with indoor and outdoor physical experiments show that our patches can seamlessly blend into the environment while successfully inducing false positive predictions at the patch location.

The remaining part of this work is arranged as follows. We first introduce related works of physical adversarial attacks and naturalistic adversarial attacks in Section II. Then, we explain our proposed attack pipeline in detail in Section III. The experimental results of our patches in digital domain and physical domain are presented in Sections IV and V,

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respectively, and finally, we conclude the whole work and introduce our future work in Section VI.

II. Related Work

Prior works include the origin of adversarial attack research, physical adversarial attacks, and naturalistic adversarial patch generation.

A. Pioneering Research of Adversarial Attacks

Deep learning technology has been widely used in many fields, but researchers have gradually found that these models are vulnerable to attacks by adversarial examples [29], [30], [31]. The adversarial attack methods can be divided into gradient-based and optimization-based methods. Fast gradient sign method (FGSM) [32], iterative FGSM (I-FGSM) [33], and projected gradient descent (PGD) [34] are all gradient-based methods that generate adversarial perturbations using the gradient information of deep models. Deepfool [35] and C&W [36] are optimization-based attack methods, which consider generating adversarial examples as an optimization problem, aiming to find the smallest perturbation that leads to false prediction results. However, applying carefully designed perturbation to the image during the camera shooting stage is difficult practically, thus limiting the application of digital adversarial attacks, and physical adversarial attack methods appear and gradually gather attention.

B. Physical Adversarial Attack

1) Physical Adversarial Attack on Natural Images: Studies have been conducted on physical adversarial attacks targeting natural images. Wu et al. [37] proposed a framework for optimizing physical realizable adversarial patches against human detectors with high transferability across different detection systems. Guesmi et al. [27] investigated an adversarial patch with the properties of an artwork, enhancing its naturalistic attribute while being able to fool the target detectors. Adversarial examples in the field of autonomous driving have been studied recently. Wang et al. [38] investigated an adversarial attack method against widely used LiDAR systems and point cloud data, to induce 3-D object detectors to misidentify a fake vehicle object by manipulating point cloud clusters of specific shapes. Sato et al. [39] realized an attack on the steering system of autonomous driving by replacing the road surface with an adversarial patch with a similar pattern. Adversarial attacks on face recognition systems are also of interest for research. Doan et al. [26] proposed a GAN-based adversarial patch for generating patches with a flower pattern; by wearing it on the hair, it can attack face recognition algorithm. Sharif et al. [40] obtained an optimized adversarial patch for face recognition scenarios with a spectacle frame shape design. Wearing the patch allows the user to be invisible to the recognition system or recognized as another person.

2) Physical Adversarial Attack on Remote Sensing Images: Due to the development of deep learning technology and the characteristics of remote sensing field in terms of multiscale, multimodality, and large data volume, deep learning models have been widely used in various remote sensing scenario-related studies. However, relatively little research has been done on the security and robustness of deep models in remote sensing field, leading to some user concerns about data security. The study of adversarial attack for remote sensing images reveals the vulnerability of deep learning models. Lian et al. [41] proposed an adaptive-patch-based framework, which uses the output of the detector as a loss function in the adversarial patch optimization process. The method achieves effective adversarial attacks in both physical dynamics and varying scales on different target objects. In the research of Huang et al. [42], multiple small-sized adversarial patches were embedded in the remote sensing image while the pattern and placement of the patches were simultaneously optimized. In this way, they minimized the difference in the image before and after applying adversarial patches and achieve the best adversarial effect. Du et al. [43] proposed an adversarial attack method for low-altitude aerial vehicle detection scenarios that avoids directly modifying the pattern of the object itself by placing patches around the target object. The method effectively realizes the attack performance of patches. In addition, they also investigated the physical-domain adversarial effects in remote sensing scenes under different weather conditions. In recent research on adversarial attacks in the remote sensing field, existing methods are relatively homogeneous in terms of optimization paradigm, usually focusing on the adversarial effect and ignoring the natural properties of patches. Meanwhile, generating a singular type of adversarial patch reduces the difficulty of defense.

C. Naturalistic Adversarial Patch Generation

Most of the existing studies on adversarial patch attack mainly focus on attack performance and seldom pays attention to the appearance of adversarial patches, which leads to the easy identification of these attacks. Therefore, researchers start to investigate the method of getting a naturalistic adversarial patch. The methods can be broadly divided into optimization-based method and generative model-based method. In optimization-based method [44], [45], [46], [47], the researchers design optimization modules that transform the style of adversarial patch into an optimization problem. In Advart [27], a cosine similarity loss is introduced to measure the visual difference between the generated adversarial patch and a given artwork. By minimizing the difference, the adversarial patch is optimized visually close to the artwork. Cheng et al. [48] employ a style transform module to constrain the style difference between the adversarial patch and a natural image. The appearance of the patch is further optimized through the use of style loss and content loss. With the continued development of generative models, researchers have found that they exhibit advantages in generating adversarial patches with naturalistic appearance. In AdvGAN [49], both the detection results from a deep classification model and a discriminator are used to jointly optimize the generator's training process, enabling the generation of adversarial examples with high perceptual quality. Kong et al. [50] proposed PhysGAN, where a GAN generates adversarial patches effective across consecutive frames while ensuring that the generated patches

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maintain visual consistency with natural images. In [25] and [26], a pretrained GAN is used to generate natural image with adversarial capability by optimizing the distribution of hidden variables in the generator's inputs. The natural appearance of adversarial patches in these methods depends on the quality of the pretrained GAN.

III. Methodology

In this section, we introduce the pipeline for optimizing adversarial patches, which are designed to deceive aerial DNN detectors by causing them to detect a nonexistent object at the patch's placement. Specifically, our proposed patches exhibit three key characteristics.

1) *Diversity:* The patch generator can map different latent vectors into adversarial patches with different patterns.
2) *Stealthiness:* The adversarial patches can visually hide in specific scenes.
3) *Physical Realizability:* Optimized patches are effective when viewed from various heights and angles, and be able to use in physical world.

A. Problem Definition

We apply adversarial patches P to the benign images by a binary mask $M \in \{0,1\}^{H\times W}$, where H and W represent the image height and width, respectively. Adversarial patches P are applied to benign aerial images through M by following the process:

$$x_{\text{adv}} = (1 - M) \odot x + M \odot P \tag{1}$$

where x and x_{adv} represent the benign image and adversarial scene image, respectively, $\odot$ represents the Hadamard product, and P represents the adversarial patches.

Previous approaches for adversarial patch generation typically optimize pixel values directly. While effective in attacking, they produce static patterns that lack adaptability to backgrounds, resulting in perceptual inconsistency against the environment, as shown in Fig. 1. In comparison, we use a generator to produce adversarial patches similar to background texture, enabling patches to blend into the surrounding environment. A combined loss is adopted during generator optimization, leading to the achievement of adversarial efficacy and patch stealthiness.

To ensure the effectiveness of adversarial patches in the physical world, the difference between the digital and physical domains is a critical factor. To address this, we adopt expectation over transformation (EoT) [51], incorporating random scale and shape transformations. We also introduce random changes in brightness, contrast, and noise during the optimization process to account for lighting variations.

B. Overview of DSAP Framework

Our proposed method can convert fixed-dimension latent vectors into adversarial patches through a trained patch generator. In our framework, during patch optimization, we extract multiple regions from benign images to form background image blocks as the source-domain image set and achieve visual stealthiness of patches by constraining the patch pattern to resemble the source-domain images. The overall training pipeline is shown in Fig. 2. After the patch generator completes training, dynamic generation of adversarial patches with different patterns can be achieved through latent space sampling. This mechanism allows the extraction of arbitrary latent vectors from a continuous latent space as input to the generator, theoretically obtaining an infinite number of adversarial patches with different patterns, as shown in Fig. 3. The characteristic possessed by the generator enables it to output diverse adversarial patterns against a single object detection model, while ensuring a tradeoff between attack performance and stealthiness of adversarial patches.

C. Source Image Extraction

Adversarial patches with similar appearance to their local placement background exhibit superior environmental blending, effectively increasing the difficulty for visual perception. To achieve this, we construct a background-related source-domain image set S by extracting patches from benign images, making the patches resemble source-domain images to be stealthy. By constraining the generated patches' appearance, we evaluate their matching degree with the local background and select the patch that is most difficult to perceive at the patch placement location, thus achieving seamless integration into the target scene.

D. Patch Location Selection

We utilize a SelectLoc function for finding suitable locations to apply patches in the input images. It identifies the patch location set $L = \{l_i\}$, where each location l_i must satisfy the following.

1) Zero overlap with existing GT objects, i.e., $\text{IoU}(l_i, GT_j) = 0 \ \forall j$. This prevents adversarial patches from covering the actual target objects already in the scene. By avoiding spatial conflicts with ground truth (GT) objects, we ensure the physical plausibility of patch placement and prevent adversarial texture optimization from the negative effect of the existing target object at the patch location.
2) Mutual exclusivity among selected patch locations, i.e., $\text{IoU}(l_i, l_k) = 0 \ \forall i \neq k$. In practical deployment, overlapping placements of patches are impossible. This constraint enforces spatial feasibility and also promotes better dispersion of adversarial patches across the target scene, ensuring the universality of adversarial attack at any position in the scene.

The SelectLoc function ensures that adversarial patches are located at physically plausible locations, while preventing occlusion conflicts that could negatively affect the optimization process.

E. Stealthy Adversarial Patch Optimization

Our method employs a GAN for generating stealthy adversarial patches. The patch generator G maps latent vectors $\mathbf{z}$ sampled from the latent space to adversarial patches $P = G(\mathbf{z})$, while the pattern discriminator D supervises the generator

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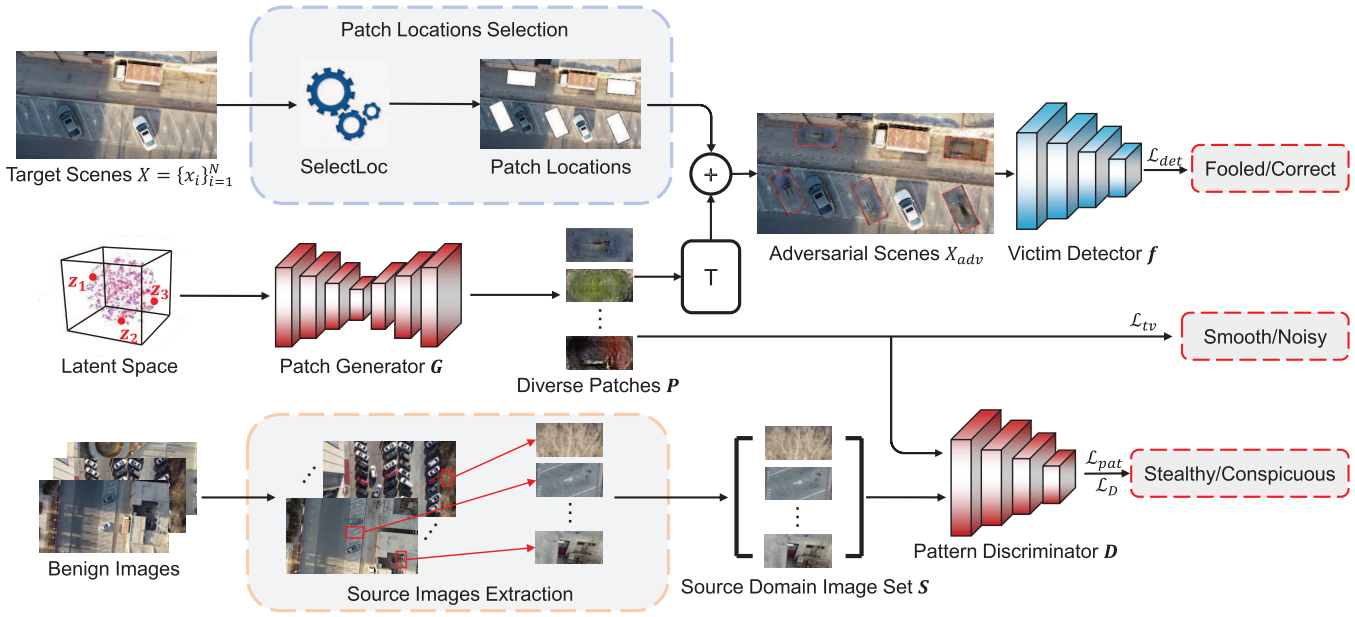


Fig. 2. Overview of our proposed DSAP pipeline. We construct a source-domain image set to constrain generated patches' appearance, making them resemble the background in the target scene. We design a SelectLoc function to select locations in the scene for physical plausible patch deployment. In adversarial attack optimization, detection loss is utilized to supervise the attack capability of generated patches.

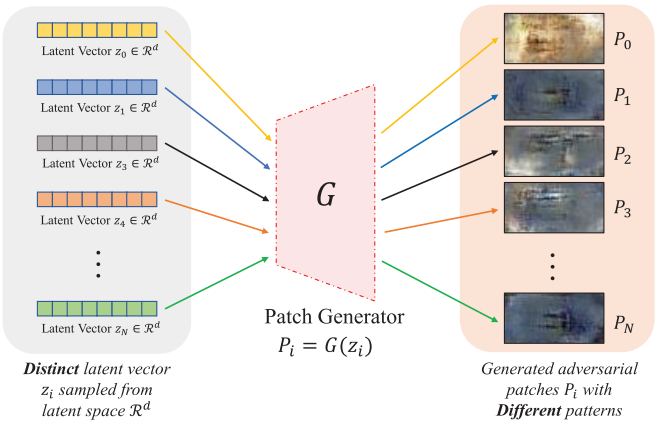


Fig. 3. Diverse adversarial patch generation. A trained generator can map distinct latent vectors to different patterns, theoretically generating an infinite number of patches with different appearance resemble source-domain images.

optimization by reducing the visual difference of generated patches and source-domain images. Assume $s \in S$, P_S is the distribution of the source-domain images, and $p(z)$ represents the latent vector distribution, the problem of generating stealthy patches can be formulated as solving

$$\min_G \max_D \mathbb{E}_{s\sim P_S}\left[D\left(s\right)\right] - \mathbb{E}_{z\sim p(z)}\left[D\left(G\left(z\right)\right)\right]. \tag{2}$$

Through this min–max optimization, G gradually learns to produce adversarial patches that conform to the distribution of S. Compared to other adversarial attack optimization methods, our proposed scene-driven stealthy pattern synthesis technique constrains the patch generation process based on source-domain images, ensuring alignment with the background environment's appearance to achieve the control of adversarial patches' visual attributes such as texture and color tone.

To optimize patch attack efficacy, patches are rendered to the selected locations to obtain the adversarial scene x_{adv}, which is later used for detector forward propagation to compute detection loss for updating the patch generator. We design the rendering pipeline incorporating transformations inspired by the EoT framework. First, random scale and shape perturbations are added to the patches. Unlike methods that focus on simple rotations or translations, we take into account a large range of distance and perspective changes when the UAV perceiving the patch, which is more suitable for an aerial imagery scenario. Second, we apply random brightness, contrast, and noise perturbations to mimic the impact of unpredictable lighting conditions and sensor imperfections in the real world.

Following white-box settings, we optimize adversarial patches by employing fake labels to guide the optimization process, thereby making the patches be recognized as a fake object. Specifically, our optimization object aims to control the victim model's perception result of an adversarial patch to a nonexisting target object, thus fooling the detector to perceive a fake object with an optimized adversarial patch.

F. Loss Functions

To achieve dual properties of attack performance and stealthiness in optimized patches, we combine adversarial loss $\mathcal{L}_{\text{D}}, \mathcal{L}_{\text{pat}}$ with detection loss $\mathcal{L}_{\text{det}}$ during adversarial training of patch generator G and pattern discriminator D. Additionally, total variation loss $\mathcal{L}_{\text{tv}}$ is utilized to ensure smoothness of adversarial patterns.

1) Detection Loss $\mathcal{L}_{det}$: To ensure adversarial efficacy, $\mathcal{L}_{\text{det}}$ encourages detector f to identify patches as spoofing objects. Let B_{GT} denotes the GT bounding boxes in input image X, y' denotes the spoofing object labels and coordinates, R is

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the rendering process, M represents adding GT masks before forward propagation, $\mathcal{J}$ is the loss function, and the detection loss calculation can be formulated as

$$\mathcal{L}_{\text{det}} = \mathcal{J}\left(f\left(M\left(R\left(X, L, P\right), B_{\text{GT}}\right)\right), y'\right) \tag{3}$$

where P denotes the adversarial patches, and L denotes the selected patch locations.

2) Adversarial Loss ($\mathcal{L}_{pat}$, $\mathcal{L}_D$): We adopt $\mathcal{L}_{\text{pat}}$ and $\mathcal{L}_{\text{D}}$ to supervise G and D optimization, respectively, during adversarial training process, ensuring the generated patch pattern resemble source-domain images. Adversarial losses are defined as

$$\mathcal{L}_{\text{D}} = \mathbb{E}_{x\sim P_S}\left[D\left(x\right)\right] - \mathbb{E}_{z\sim p(z)}\left[D\left(G\left(z\right)\right)\right] \tag{4}$$

$$\mathcal{L}_{\text{pat}} = -\mathbb{E}_{z\sim p(z)}\left[D\left(G\left(z\right)\right)\right] \tag{5}$$

where P_S is the distribution of source-domain images, and $p(z)$ is the latent vector prior.

3) Total Variation Loss $\mathcal{L}_{tv}$: To encourage spatial smoothness and reduce high-frequency component, $\mathcal{L}_{\text{tv}}$ is used to constrain generator optimization, which can be expressed as

$$\mathcal{L}_{\text{tv}} = \sum_{i,j} \sqrt{\left(q_{i,j} - q_{i+1,j}\right)^2 + \left(q_{i,j} - q_{i,j+1}\right)^2} \tag{6}$$

where $q_{i,j}$ is the pixel value on patch P.

Finally, the generator G, which can produce adversarial patterns, is optimized via the combined loss

$$\mathcal{L}_{\text{G}} = \alpha \cdot \mathcal{L}_{\text{det}} + \beta \cdot \mathcal{L}_{\text{pat}} + \gamma \cdot \mathcal{L}_{\text{tv}} \tag{7}$$

with α,β, and γ balancing the adversarial strength, patch pattern, and spatial smoothness. Meanwhile, D is updated solely by $\mathcal{L}_D$ to refine the pattern discriminator. Our proposed DSAP optimization method is summarized in Algorithm 1. Given benign aerial images X and latent vectors $\mathbf{z}$, our algorithm generates adversarial patches by alternately optimizing G and D. Our method achieves adversarial pattern control. Compared to other methods, it balances the tradeoff between attack efficacy and stealthiness, and can synthesize diverse adversarial patches of different visual appearances without multiple trainings.

Algorithm 1 Deployable Scene-Adaptive Adversarial Patch Optimization

Input : benign aerial images X, latent vectors $\mathbf{z} \sim p(\mathbf{z})$, source domain images set S, initialized G (weights θ_G) and D (weights θ_D), G update times G_{iters}, D update times D_{iters}, generator learning rate η_G, discriminator learning rate η_D, patch placement selection function SelectLoc$(\cdot)$, adding masks function $\mathcal{M}(\cdot)$, rendering function $\mathcal{R}(\cdot)$, detector $f(\cdot)$, maximum iteration N.

Output : Adversarial patches P.

Initialize G and D;
for $i = 1$ to N **do**:
  for $j = 1$ to D_{iters} **do**:
    Sample batch $\{s_i\}_{i=1}^{n} \in S$;
    Sample $\mathbf{z} \sim p(\mathbf{z})$;
    Calculate $\mathcal{L}_{\text{D}} = -\mathbb{E}_{z\sim p(z)}[G(\mathbf{z})]$;
    Update D: $\theta_D \leftarrow \theta_D - \eta_D \cdot \nabla_{\theta_D}\mathcal{L}_D$;
  end for
  for $k = 1$ to G_{iters} **do**:
    Sample $z \sim p(z)$, generate patches $P = G(z)$;
    Calculate $\mathcal{L}_{\text{pat}} = -\mathbb{E}_{z\sim p(z)}[D(G(z))]$;
    Sample $x \in X$, coordinate GT boxes B_{GT};
    Select patch location $L = \text{SelectLoc}(x)$;
    Render adversarial scene $x_{\text{adv}} = \mathcal{R}(X, L, P)$;
    Apply GT masks to x, obtain $\mathcal{M}(x_{\text{adv}})$;
    Forward propagation, $f(\mathcal{M}(x_{\text{adv}}))$;
    Calculate detection loss $\mathcal{L}_{\text{det}}$ according to Eq. 3;
    Calculate TV loss $\mathcal{L}_{\text{tv}}$ according to Eq. 6;
    Calculate $\mathcal{L}_{\text{G}}$ according to Eq. 7
    Update G: $\theta_G \leftarrow \theta_G - \eta_G \cdot \nabla_{\theta_G}\mathcal{L}_{\text{G}}$
  end for
end for
Sample $z \sim p(z)$ and generate patches $P = G(z)$
Return P

IV. Digital Attack Experiments

In this section, we conduct digital attack experiments to validate the effectiveness of the proposed DSAP attack. We first describe the experimental settings, followed by a comparative analysis of attack performance, patch stealthiness, and patch diversity. Defense difficulty study, ablation studies, and more discussions are also further presented in this section.

A. Experimental Setup

1) Dataset: We establish our experiments using three datasets, including the following.

1) *UAV-ROD dataset* [52] is an aerial vehicle object detection dataset captured by a UAV. It contains 1577 aerial images with 30090 rotated bounding box annotations for vehicle instances. The images are acquired at 30–80-m altitudes under a top–down perspective, covering diverse scenarios, e.g., urban roads and parking lots, where each scenario corresponds to a sequence of consecutive frames.
2) *DOTA dataset* [53] contains 2806 high-resolution aerial images with 15 categories of targets in total. To validate the DSAP performance of creating different spoofing objects, we selected four typical classes (plane, small vehicle, large vehicle, and ship) in the DOTA dataset for attack research.
3) *Custom street scene dataset*, comprising 550 aerial images with camera resolution 5280 × 3956 captured by a DJI Mavic 3 Pro UAV. This dataset is specifically designed to transfer digital adversarial patches to physical domain, where the data serve as simulating real-world deployment scenarios for patch training.

2) Evaluation Metric: Attack success rate (ASR) measures the effectiveness of adversarial attacks by evaluating the percentage of successfully detected spoofing object on the patch location. We consider a successful attack to satisfy two conditions: 1) confidence of spoofed object exceeds a predefined threshold and 2) predicted bounding box aligns sufficiently with predefined target region. To quantify the stealthiness

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of adversarial patches, we employ learned perceptual image patch similarity (LPIPS) [54] and structure similarity index measure (SSIM) [55], which indicate perceptual difference and low-level image feature discrepancy after embedding patches, respectively. A lower LPIPS and a higher SSIM value indicate higher perceptual similarity, implying that the adversarial patch blends seamlessly into the background without introducing noticeable artifacts. To quantify patch diversity, we input the patches to an ImageNet-pretrained VGG-16 classifier and extract penultimate-layer logits prior to SoftMax layer as output. We define mean interpatch similarity (MIPS) as the average cosine similarity between output logits of patches, with a lower value indicating higher diversity.

3) Implementation Details: Six rotated object detection models including Rotated RetinaNet (Retinanet-o) [56], Rotated FasterRCNN (Faster RCNN-o) [57], Gliding Vertex [58], Oriented R-CNN [59], RoI Transformer [60], and S^2A-Net [61] are selected as our victim models. All detectors utilize ResNet-50 backbones pretrained on ImageNet and are first trained on the UAV-ROD dataset following their official configuration. For adversarial patch generation, generator takes 100-D latent vectors as input and generate 32×64 pixels adversarial patches, while the discriminator takes 32×64 pixel patches and images in the source-domain image set as input. During training process, the discriminator is updated for ten times while the generator is updated once in each iteration, ensuring that the discriminator provides reliable guidance for the generator's learning process. α, β, and γ in (7) are set to {0.1, 1.0, 0.1}, respectively, during adversarial patch optimization. For each scene image, ten patches are applied to appropriate locations, and the optimization process lasts for 800 epochs with a batch size of 8. We employ the RMSprop optimizer for both generator and discriminator with the same learning rate of 0.0002, balancing convergence speed and training stability. The experiments can be finished on a single NVIDIA RTX 3090 GPU with 24-GB RAM.

B. Attack Performance of DSAP

Our proposed adversarial patches can effectively attack aerial object detectors, as shown in Fig. 4. Specifically, when embedded into the scene, these patches induce the detection algorithm to falsely identify adversarial patches as spoofing objects, thereby affecting aerial imagery perception. To validate the effectiveness of our approach, we conduct comparison experiments with the following attacking methods.

1) *Random:* Nonoptimized patches with pixel values sampled from a uniform distribution, serving as a control set.
2) *I-FGSM:* Adversarial noise iteratively added to target regions along the image gradient direction to trigger detection results with spoofing objects.
3) *AdvPatch [13]:* Ordinary adversarial patch generation that directly performs pixel-level optimization.
4) *AdvART [27]:* Utilizing pixel-level constrain during optimization to make patches effective while maintain the appearance of a given artwork.
5) *NAP [25]:* Searching for a suitable latent vector in the latent space and use a pretrained generator to map it into an adversarial patch with naturalistic appearance.

Notably, all attack methods are evaluated on the same test set using identical ASR setting to determine attack performance, with perturbations or patches standardized to consistent sizes. We showcase the visual comparisons of these methods in Fig. 5.

We evaluated the patch attack performance on a sequence of 50 consecutive frames. For each frame, two nonoverlapping locations were randomly selected. Over 50 iterations, this process generated 5000 adversarial attacks. The ASR, calculated as the ratio of successful attacks to total attempts, was adopted as the performance metric. Specifically, increasing confidence thresholds imposed stricter criteria for successful attacks, requiring higher erroneous detection confidence from the victim detectors. The results are shown in Table I. First, the low ASR of nonoptimized random patches across all detectors demonstrated that merely embedding patches into scenes does not inherently trigger false positive detection results. Effective adversarial attacks require deliberate adversarial pattern optimization. Second, all these methods of optimization have achieved high adversarial performance, and the DSAP has good attack performance under varying confidence thresholds. For example, under the strictest confidence threshold of 0.7, our method can achieve an ASR of over 90%. While DSAP exhibits a slightly lower ASR than AdvPatch and AdvART, its generator-guided optimization paradigm enables controllable adversarial pattern generation, thereby significantly enhancing stealthiness compared to these pixel-level optimization. This strategic tradeoff between attack performance and adversarial pattern control demonstrates higher practical value in real-world adversarial attack deployment situation where stealthiness is critical. Third, our adversarial patches can achieve high attack performance on various detectors. At the situation the confidence threshold set to 0.5, our adversarial patches achieved over 90% ASR on different detectors, demonstrating the generalization of our proposed adversarial attack method in the face of different models.

C. Stealthiness of DSAP

To assess the stealthiness of adversarial patches in the scene, we develop a quantitative evaluation employing SSIM and LPIPS metrics computed from scene images before and after patch embeddings. Specifically, we adopt a test set containing 50 consecutive scene images. In each frame, four spatially nonoverlapping locations avoiding GT regions are selected. After embedding optimal stealthy patches based on local texture similarity, we extract fixed-size image blocks from both patch-embedded regions in the adversarial scene and their corresponding areas in benign images, constructing a set of embedded patch regions and a reference set of original background regions. The quantitative metrics between these two sets are calculated to evaluate visual perception changes caused by embedded patches.

Stealthiness comparison of different attack methods is shown in Table II. Among all compared adversarial attacks,

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Fig. 4. Visualization of DSAP attacks on single-category vehicle objects and multicategory objects: (a) attack results on the vehicle detection dataset UAV-ROD in the first row; (b) attack results on our custom vehicle detection dataset in the second row; and (c) attack results on the DOTA dataset in the third row. The locations for deploying patches are indicated by bright pink arrows.

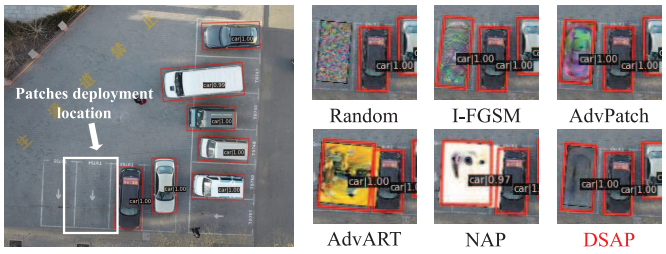


Fig. 5. Different adversarial attack methods compared at the same location, with attack visualization and corresponding detection results presented. The DSAP melts into the surrounding environment and thus achieves outstanding visual stealthiness compared to other methods.

our DSAP achieves the lowest LPIPS metric, demonstrating superior visual stealthiness, with minimal perceptual discrepancy from the original scenes after patch embedding. This stems from the resemblance of adversarial patterns and background texture. In contrast, AdvPatch, AdvART, and NAP exhibit a high LPIPS value and low SSIM score, indicating poor stealthiness. The reason is that its optimization solely focuses on attack performance, thus obtaining background-agnostic patterns with significant divergence in both high-level semantics and low-level features. Moreover, DSAP has a lower LPIPS value and higher SSIM score compared to the I-FGSM attack method, which is attributed to their different perturbation mechanism: I-FGSM employs pixelwise perturbation while DSAP implements stealthy patch embedding. Although DSAP does not achieve optimal performance under the SSIM metric for measuring low-level structural similarity, it has good performance at the high-level semantic feature. The visualization results in Fig. 5 further confirm that DSAP has a better stealthiness.

D. Diversity of DSAP

DSAP can offer a high pattern diversity through latent space sampling. We compare the diversity of AdvART, AdvPatch, and NAP through a quantitative experiment. For comparison methods, distinct visual appearances are generated by using varying random seeds during training. For DSAP, diverse adversarial patches are generated by mapping random sampled latent vectors. We show diverse patches of DSAP in Fig. 6, calculate MIPS of different adversarial patches, and report the results in Table III. The experimental results demonstrate that NAP and DSAP generate adversarial patches with superior visual variation compared to AdvART and AdvPatch. Pixel-level optimization methods, constrained by similar random noise initialization conditions and the same optimization objectives, lead to similar visual appearance. In contrast, GAN-based methods leverage a generator to map diverse distribution latent vectors into visually different adversarial textures. Compared to NAP, DSAP can obtain diverse adversarial patches through a single forward propagation, avoiding the need for repeated training based on different random seeds.

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TABLE I
ASR (%) ON UAV-ROD DATASET OF DIFFERENT VICTIM DETECTORS

Confidence	Method	Stealthy	Universal	Victim Detectors D1	D2	D3	D4	D5	D6
0.1	Random	-	-	5.80	33.46	9.98	5.36	22.80	28.10
	I-FGSM	✗	✗	95.26	95.36	89.02	94.60	94.90	97.70
	AdvPatch [13]	✗	✓	99.52	99.56	99.44	100.00	99.88	99.94
	AdvART [27]	✗	✓	100.00	99.00	96.68	99.90	98.30	100.00
	NAP [25]	✗	✓	98.66	97.28	98.18	89.16	99.78	99.62
	DSAP	✓	✓	96.52	98.36	98.16	98.04	99.22	99.70
0.3	Random	-	-	2.96	28.40	7.72	3.82	17.72	10.14
	I-FGSM	✗	✗	93.24	94.64	88.08	93.96	94.42	96.52
	AdvPatch [13]	✗	✓	99.50	99.32	99.54	99.98	99.92	99.92
	AdvART [27]	✗	✓	100.00	98.68	95.46	99.90	97.46	100.00
	NAP [25]	✗	✓	93.50	96.76	98.16	85.28	99.56	99.24
	DSAP	✓	✓	93.72	98.06	98.20	97.98	98.96	99.60
0.5	Random	-	-	1.22	26.36	6.18	3.18	14.14	3.68
	I-FGSM	✗	✗	91.54	94.08	87.34	93.56	93.64	95.22
	AdvPatch [13]	✗	✓	99.60	99.56	99.32	99.94	99.88	99.96
	AdvART [27]	✗	✓	99.86	98.48	94.82	99.66	96.18	99.92
	NAP [25]	✗	✓	85.02	96.32	97.60	81.58	99.26	97.66
	DSAP	✓	✓	92.78	97.92	97.66	97.52	98.86	99.08
0.7	Random	-	-	0.70	21.74	5.22	2.36	11.32	0.74
	I-FGSM	✗	✗	89.68	93.58	86.64	93.18	92.90	92.64
	AdvPatch [13]	✗	✓	99.60	99.48	99.42	99.94	99.84	99.92
	AdvART [27]	✗	✓	99.62	98.02	94.26	99.60	94.70	99.12
	NAP [25]	✗	✓	67.84	95.82	97.16	78.62	98.68	89.44
	DSAP	✓	✓	90.36	97.54	97.18	97.00	98.46	98.62

[1] D1, D2, D3, D4, D5, D6 represents Retinanet-o, Faster-RCNN-o, Gliding Vertex, Oriented RCNN, RoI Transformer and S^2A-Net respectively.
[2] 'Stealthy' indicates whether adversarial patches perceive visual stealthiness, 'universal' indicates whether adversarial patches are universal across different locations and scenes.

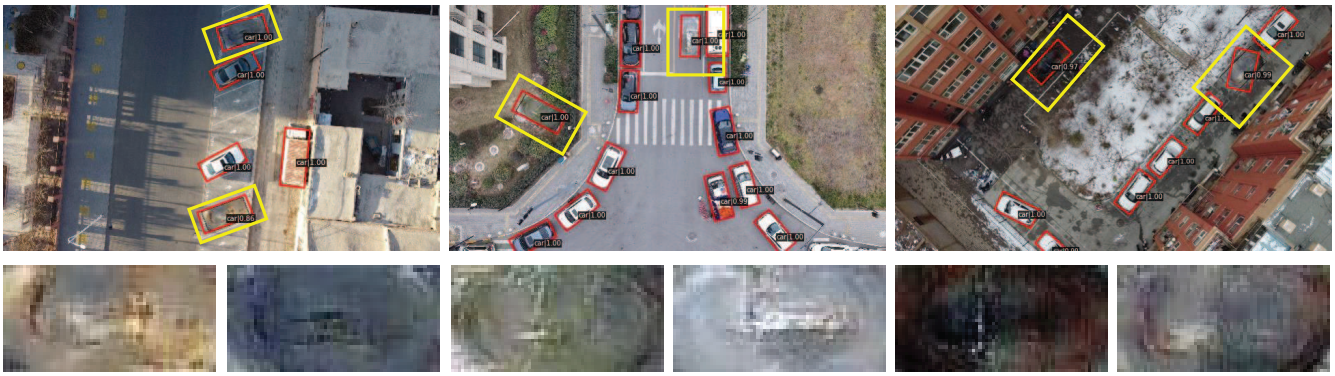


Fig. 6. Visualization of diverse adversarial patches deployed at different locations within the scenes, which exhibit different appearances and attack ability by mapping latent vectors with different distributions through a trained patch generator.

E. Difficulty on Defense

To validate the defense difficulty of different adversarial attacks, we evaluate four methodologies including two pixel-level defense, a semantic-level defense, and a model enhancement defense. For pixel-level defense, nonlocal mean denoising and JPEG compression are used; the former suppresses local noise introduced by adversarial patches, while the latter eliminates high-frequency components across the full image. For semantic-level defense, a lightweight segment network is used for adversarial region localization and patch elimination. For model enhancement defense, we enhance detector robustness through adversarial example injection during fine-tuning.

Defense results are shown in Fig. 7, which demonstrate that both pixel-level defense methods exhibit negligible defensive capabilities against AdvPatch and DSAP, with some cases even exhibit negative defensive efficacy where ASR decrease < 0, as they primarily manipulate low-level image features, whereas adversarial patches leverage high-level adversarial patterns semantically aligned with target objects, making them robust to low-level feature variations.

Semantic-level and model enhancement defense methods show significant defense performance, effectively suppressing the ASR of AdvPatch. However, when facing DSAP attacks, these two defense methods exhibit only partial protective efficacy, as the ASR remains relatively high after defense. This stems from the dependence of data-driven defense methods

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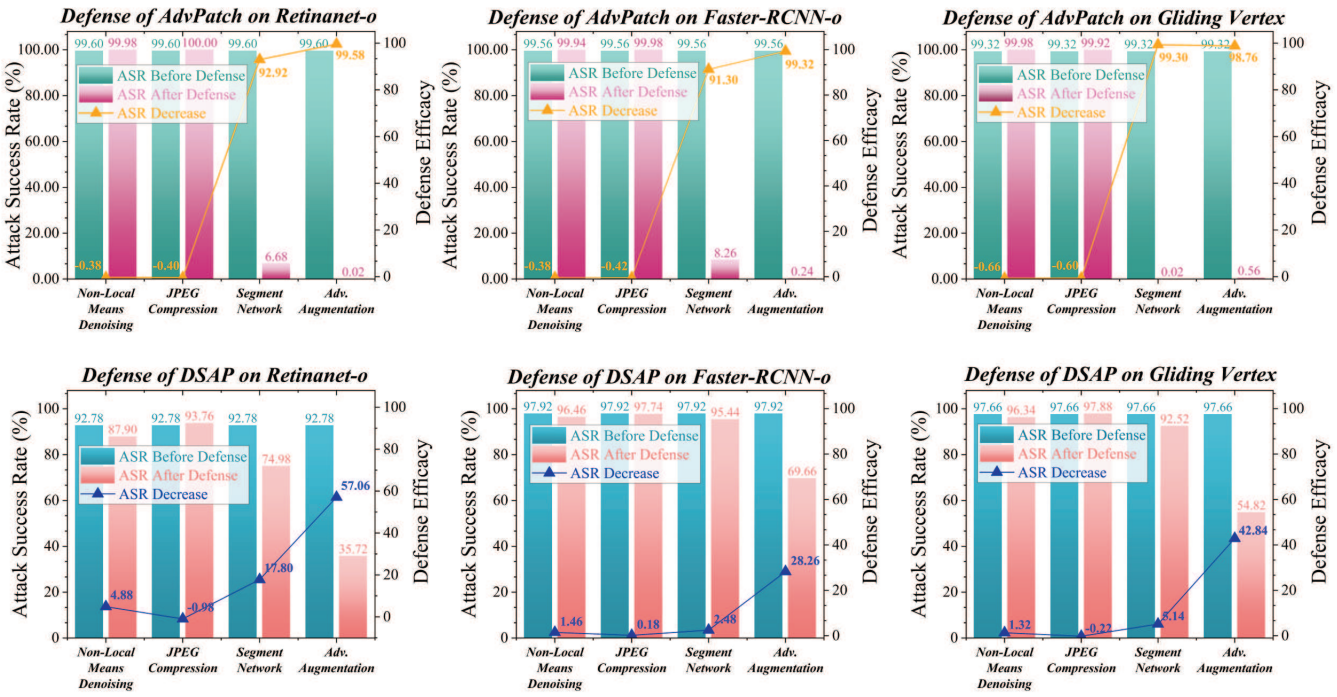


Fig. 7. Comparative effectiveness of different defensive strategies against AdvPatch and DSAP. Bar charts present the ASR before and after implementing defense measures, while line charts quantify the ASR decrease caused by defenses. A low line value indicates diminished defensive efficacy.

TABLE II
COMPARISON OF STEALTHINESS ON DIFFERENT ATTACK METHODS

Methods	Metrics	Detectors D1	D2	D3	D4	D5	D6
I-FGSM	SSIM↑	0.5955	0.5954	0.5978	0.6002	0.5983	0.5923
	LPIPS↓	0.5727	0.5719	0.5725	0.5720	0.5787	0.5923
AdvPatch	SSIM↑	0.4511	0.4709	0.4727	0.4568	0.4546	0.4480
	LPIPS↓	0.5609	0.5462	0.5298	0.5488	0.5518	0.5501
AdvART	SSIM↑	0.3451	0.3410	0.3429	0.3353	0.3394	0.3444
	LPIPS↓	0.7185	0.7277	0.7335	0.7121	0.7222	0.7162
NAP	SSIM↑	0.5078	0.5104	0.4168	0.4968	0.5162	0.5200
	LPIPS↓	0.7815	0.8000	0.7160	0.7977	0.8071	0.8047
DSAP	SSIM↑	0.5470	0.5414	0.5348	0.5266	0.5339	0.5440
	LPIPS↓	0.4236	0.4242	0.4470	0.4462	0.4415	0.4319

[1] SSIM quantifies the structural similarity of images before and after patch applying at the low-level feature, while LPIPS reflects the perceived similarity of benign images and adversarial scenes at the high-level semantic level.
[2] D1, D2, D3, D4, D5, D6 represents Retinanet-o, Faster-RCNN-o, Gliding Vertex, Oriented RCNN, RoI Transformer and S^2A-Net respectively.

TABLE III
DIVERSITY OF DIFFERENT ADVERSARIAL PATCHES

Methods	AdvART	AdvPatch	DSAP	NAP
MIPS↓	0.9252	0.8953	0.7725	0.6449

on training data distributions—they establish effective protection solely within the observed adversarial pattern space, while showing marked generalization capability decrease on unknown adversarial patterns beyond the training data distribution.Our proposed method can generate diverse adversarial patterns that occupy unexplored attack spaces unobserved during defensive training, thereby being easier to escape from defense systems compared to ordinary attacks that only employ simple adversarial patterns.

F. Transferability

We investigate the transferability of patches against convolutional neural network based (CNN-based) and Transformer-based detectors. Results are shown in Table IV. As for intraarchitecture transferability, the DSAP attack demonstrates strong transferability among CNN-based detectors, while exhibiting significant fluctuation across Transformer-based detectors. This stems from internal structural differences of detectors. CNN-based models have simpler structure and smoother decision boundaries than Transformer-based models, and thus can easily learn universal adversarial feature shared across models. In contrast, Transformer-based detectors have more complex decision boundaries and hinder the optimization process from stably learning universal adversarial features. As for cross-architecture transferability, attack performances on models with different structures are inferior to that sharing the same backbone. This discrepancy stems from different feature extraction mechanisms: CNN-based models focus on local features around target objects during detection, while Transformer-based models incorporate global contextual information. The divergence leads to different adversarial texture optimization direction, thus showing relatively weak transferability.

G. Ablation Study

To investigate the properties of our proposed DSAP, we conduct ablation studies on the weight combination in (7), the

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TABLE IV

TRANSFERABILITY OF OUR PROPOSED ADVERSARIAL PATCHES ACROSS DIFFERENT DETECTORS

Src / Vic	D1-Res50	D2-Res50	D3-Res50	D4-Res50	D5-Res50	D6-Res50	D1-SwinT	D2-SwinT	D3-SwinT	D4-SwinT	D5-SwinT	D6-SwinT
D1-Res50	92.36	92.06	91.88	95.58	95.66	96.26	73.54	76.44	77.56	93.64	93.56	85.14
D2-Res50	96.90	98.06	97.42	98.74	98.82	98.84	88.88	88.66	89.22	98.32	98.14	93.68
D3-Res50	96.16	96.88	97.64	98.60	98.36	98.10	84.56	88.56	88.64	97.80	98.56	94.44
D4-Res50	90.48	92.18	93.40	96.98	96.60	94.32	76.30	79.80	78.72	95.88	96.28	87.40
D5-Res50	95.64	96.14	96.42	98.40	98.70	97.96	82.54	87.44	86.18	97.88	97.94	92.30
D6-Res50	97.50	97.50	97.60	99.08	98.74	99.18	88.60	90.36	89.88	98.58	98.80	96.30
D1-SwinT	89.00	88.70	88.26	93.24	93.52	94.36	86.98	86.08	88.04	96.04	96.38	93.70
D2-SwinT	93.52	94.48	94.25	97.22	97.64	97.56	98.52	94.52	95.22	96.86	97.22	97.48
D3-SwinT	89.88	91.54	92.06	95.04	95.46	94.34	87.68	90.64	92.68	97.74	97.82	95.24
D4-SwinT	75.82	76.74	78.66	85.48	84.46	84.82	66.90	72.54	71.68	93.10	94.12	83.26
D5-SwinT	74.10	74.90	76.18	81.86	81.32	78.60	64.82	67.98	67.04	93.58	94.44	78.20
D6-SwinT	90.66	90.92	90.98	94.00	95.28	95.22	85.62	88.64	90.18	97.26	97.54	96.30

[1] Adversarial patches are trained on source detectors (Src) and evaluated against unknown victim detectors (Vic) under black-box conditions. Grey shaded cells represent the white-box attack results, while others represent black-box results.
[2] D1, D2, D3, D4, D5, D6 represents Retinanet-o, Faster-RCNN-o, Gliding Vertex, Oriented RCNN, RoI Transformer and S^2A-Net respectively.

TABLE V

EFFECTS OF ADVERSARIAL PATCH NUMBER PER IMAGE DURING TRAINING ON ATTACK PERFORMANCE (ASR, %) AND STEALTHINESS

Patch Num	1	2	4	8
ASR↑	99.82	99.88	93.18	91.26
LPIPS↓	0.5267	0.5567	0.4661	0.4493
Patch Num	16	32	64	128
ASR↑	89.02	92.48	92.76	92.64
LPIPS↓	0.4233	0.4290	0.4204	0.4235

[1] ASR are reported at confidence threshold of 0.5.

TABLE VI

EFFECTS OF PATCH SIZE ON TRADEOFF BETWEEN ATTACK PERFORMANCE (ASR, %) AND STEALTHINESS

Scale Factor	0.5	0.7	1.0	1.5
ASR↑	12.14	74.90	92.78	95.36
LPIPS↓	0.1669	0.2878	0.4236	0.4706

[1] ASR are reported at confidence threshold of 0.5.

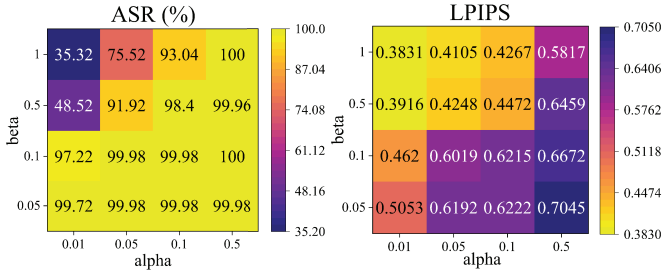


Fig. 8. Tradeoff between adversarial efficacy and stealthiness, and yellow-dominated color indicates optimal performance. ASRs are reported at a 0.5 confidence threshold.

generator loss function, and the number of embedding scene patches during an iteration.

1) Number of Patches per Image During Training: The number of patches placed within a batch of scene images in a single iteration affects the final performance of patch generation. We optimized the generator under different conditions of patch numbers and evaluated the quantitative performance of the generated patches. The experimental results are shown in Table V. When the patch number in single iteration is too small, the optimization of discriminator fails, which leads to significant difference between generated patches and source-domain images. The generator optimizes primarily for adversarial efficacy results in patches achieving an ASR over 99%, while suffering a low stealthiness level (high LPIPS value). As the number of patches increases, the discriminator gradually learns to distinguish generated patterns from source-domain images, and thus, generated patches become more stealthy and the generator achieves a balance between adversarial efficacy and stealthiness.

2) Size of Deployed Patches: The size of patches can affect attack performance and stealthiness. In our experiment, we choose scale factors {0.5, 0.7, 1.0, 1.5} to scale the adversarial patches before deploying them in the target scene, and quantitatively evaluate ASR and stealthiness. Results in Table VI indicate that increasing patch size enhances attack efficacy but concurrently elevates human perception level, thereby reducing their stealthiness. This phenomenon is intuitive: larger patches exhibit clearer adversarial textures that are more easily detected by victim models, while becoming more noticeable for human eyes.

3) Weight Combination in Generator Loss Function: Through the evaluation on different weight combinations, we quantify the adversarial effectiveness and stealthiness of the embedded patches, with results shown in Fig. 8. The experimental results show that the weight combination significantly impacts the tradeoff between adversarial efficacy and stealthiness. To maintain the tradeoff between adversarial and stealthy performance, the ratio between α and β in (7) must be properly balanced (experiments indicate that the optimal

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TABLE VII
DETECTION PERFORMANCE OF PRETRAINED DETECTORS ON DOTA DATASET

Model Type	Model Name	Backbone	mAP(%)
One-stage	Retinanet-o	ResNet50	68.42
	S^2A-Net	ResNet50	73.91
Two-stage	Faster-RCNN-o	ResNet50	73.40
	Gliding Vertex	ResNet50	73.23
	Oriented RCNN	ResNet50	75.69
	RoI Transformer	ResNet50	76.08

ratio is approximately 0.1:1). When α is too large, the patches exhibit high adversarial performance on the detector, but show a significant difference in pattern compared to source-domain images. Conversely, when α is too small, generated patches have high pattern similarity to the source images but exhibit poor attack performance.

During training, the generator and discriminator maintain a dynamic balance through competition; thus, the optimization failure of either module disrupts this competitive relationship, resulting in the simultaneous degradation of both networks. When the detection loss $\mathcal{L}_{\text{det}}$ dominates, the generator primarily optimizes toward increasing the attack performance of the patches, potentially converging to a local optimum. This leads to the failure optimization of discriminator, which becomes unable to effectively distinguish between the source-domain images and the patches, and lose its supervisory role in guiding the generator's optimization. Meanwhile, when the weight of $\mathcal{L}_{\text{det}}$ is too small, the generator fails to learn sufficient texture information to induce false detection, resulting in poor adversarial effectiveness of the generated patches.

H. Attacks on DOTA Dataset

We further evaluate our method against multicategory objects on the DOTA dataset. Following the official implementation of the MMRotate framework [62], we employ six pretrained detectors to optimize adversarial patches. The settings along with detection performances of these detectors are shown in Table VII. During optimization, we extract image blocks from DOTA to form a source-domain image set, constraining adversarial patterns to resemble the background. Notably, we adopt a target-adaptive patch sizing strategy to ensure that the patch size matches corresponding existing GT object size.

The attack performance shown in Table VIII, along with visualization shown in Fig. 4, demonstrates that our method has been successfully extended to the DOTA dataset, achieving effective multicategory adversarial patch generation. Among the four target classes, adversarial patches of plane exhibit the best attack efficacy, while others show limited attack performance. This phenomenon can be attributed to two factors. First, patches designed for plane occupy a larger area on images compared to that of other targets, which enables more perturbation by applying adversarial patches. Second, the differences in size and intraclass morphology of planes are smaller than those of large vehicles, small vehicles, and ships, making it easier for adversarial patterns to obtain a universal feature with attack ability. In contrast, other classes exhibit intraclass variation in both scale and morphology, coupled with limited patch coverage areas, making it difficult to effectively learn universal adversarial pattern, ultimately affecting the attack performance of adversarial patches.

TABLE VIII
ASRS OF OUR PROPOSED DSAP ATTACK ON DOTA DATASET

Class	Type	Detectors					
		D1	D2	D3	D4	D5	D6
PL	Random	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	DSAP	0.9174	0.8644	0.8042	0.9578	0.8884	0.8892
LV	Random	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	DSAP	0.5600	0.6166	0.4830	0.5734	0.6144	0.4706
SV	Random	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	DSAP	0.5364	0.4702	0.5356	0.5376	0.4646	0.6100
SH	Random	0.0003	0.2454	0.1212	0.0856	0.0024	0.0188
	DSAP	0.5426	0.7324	0.7692	0.7774	0.8846	0.9258

[1] Attack success rates at 0.3 confidence threshold are reported.
[2] D1, D2, D3, D4, D5, D6 represents Retinanet-o, Faster-RCNN-o, Gliding Vertex, Oriented RCNN, RoI Transformer and S^2A-Net respectively.

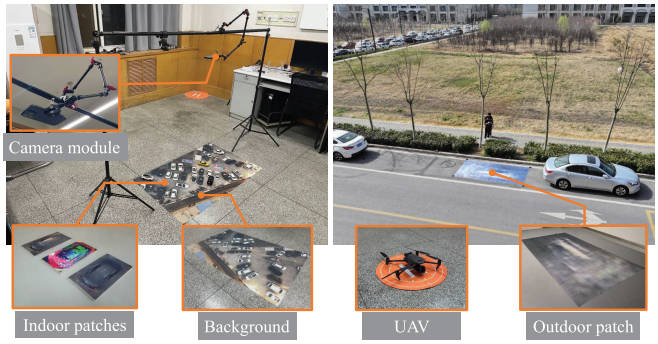


Fig. 9. (Left) Indoor aerial physical adversarial attack simulation platform and (right) outdoor physical adversarial attack platform.

TABLE IX
ROBUSTNESS OF PHYSICAL OUTDOOR ADVERSARIAL ATTACK (PHYSASR, %) ON VARIED HEIGHTS AND VIEWING ANGLES

Patch Type	Height (angle=90°)				Angle (height=30m)		
	25m	30m	35m	40m	90°	75°	60°
DSAP@Road	37.42	96.89	100.00	89.74	96.89	100.00	100.00
DSAP@Grass	56.19	98.93	91.90	82.73	98.93	90.63	88.02

[1] 'Height' denotes the UAV flying altitude, while 'Angle' represents the angle between camera shooting direction and horizontal plane.
[2] Two different adversarial patches with similar pattern to road surface (DSAP@Road) and grass (DSAP@Grass) are adopted for robustness evaluation.

TABLE X
PHYSASR (%) OF VARIOUS ENVIRONMENTAL LIGHTING CONDITIONS AND UAV MOTION STATES

Patch Type	Timing		Motion State		
	Afternoon	Dusk	Horizontal Translation	Vertical Translation	Rotation
DSAP@Road	82.55	35.57	82.55	42.28	56.37

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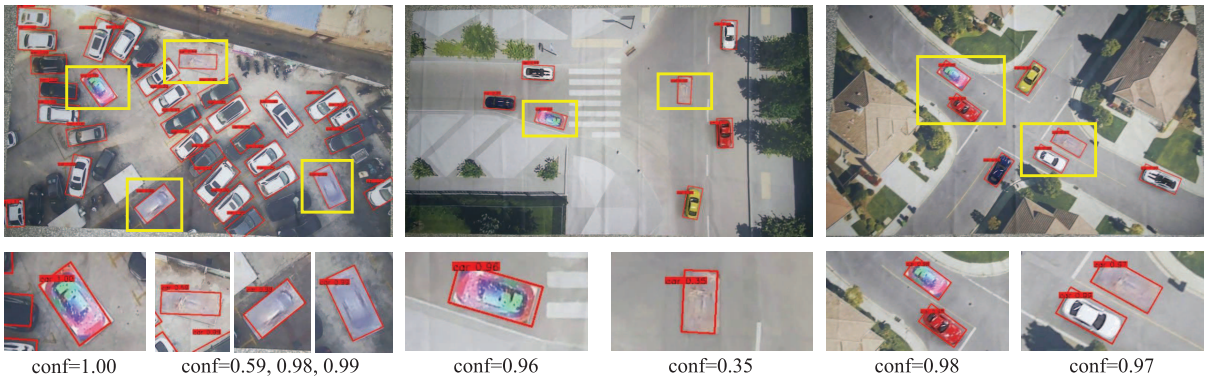


Fig. 10. Visualize comparison of AdvPatch and DSAP in indoor platform. DSAP achieves better stealthiness compared to AdvPatch.

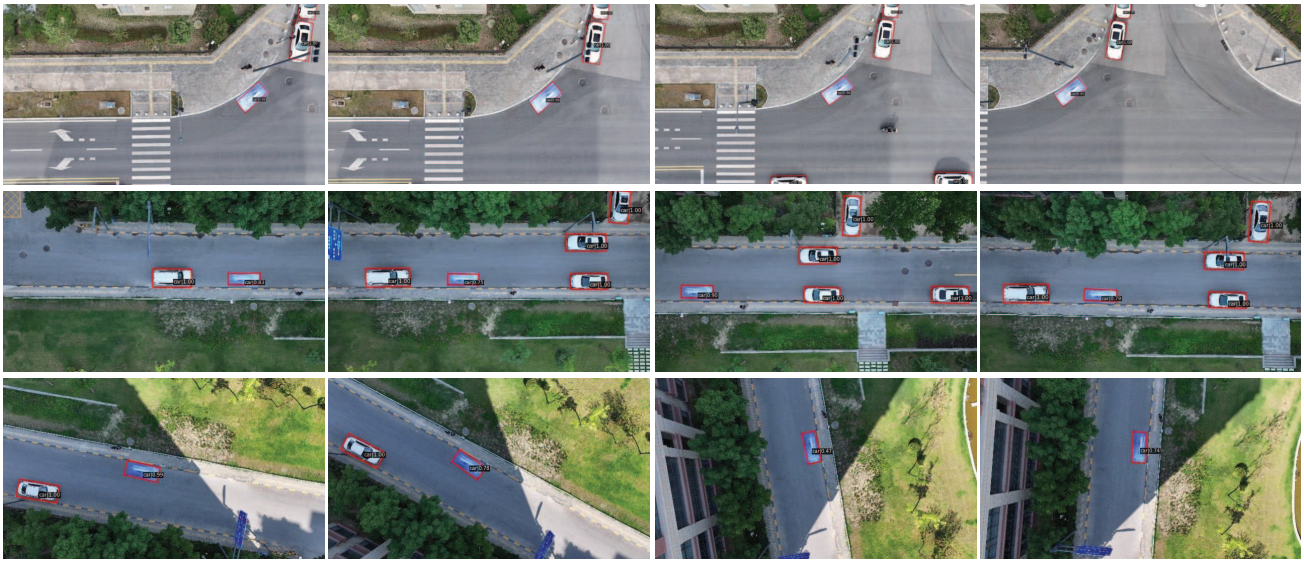


Fig. 11. Visualization of our proposed adversarial patches deployed in an outdoor scenario. The first and second rows display images captured during the afternoon and at dusk, respectively, while the third row shows images captured under UAV rotation motion.

V. Physical Attack Experiments

In this section, we conduct an adversarial attack targeting a real-world aerial vehicle detection system. We establish physical experimental settings, followed by a comprehensive evaluation of the patches' attack efficacy. Experiments across different altitudes, viewing angles, environmental lighting conditions, and UAV motion stats are also adopted to show the practical deployment of our attack method.

A. Implementation Details

We established an indoor physical adversarial attack simulation platform for UAV, simulating the real world at a 1:36 scale ratio, which is illustrated in Fig. 9 (left). There are three core components in our indoor platform.

1) *Background:* Top–down view backgrounds to simulate aerial observation scenario.
2) *Indoor Adversarial Patches:* Printed patches are deployed in the background to achieve physical adversarial attack.
3) *Camera Module:* A camera is mounted above ground to simulate real-world UAV aerial photography.

Meanwhile, we construct an outdoor adversarial attack platform using DJI Mavic 3 Pro with a camera resolution of 5280 $\times$ 3.956, which is shown in Fig. 9 (right). Image acquisition is performed through a bird's-eye view at 30-m altitude, where captured street view data are subsequently processed by an offline-deployed RetinaNet-o vehicle object detector. The adversarial patch is trained in the digital domain at 32 $\times$ 64 pixel resolution and then resized to 320 $\times$ 640 pixels through bilinear interpolation before physically printed into a 2 $\times$ 4 m adversarial patch. Attack effectiveness is quantified using physical ASR (PhysASR) as the evaluation metric, which measures the proportion of frames where the patch successfully fools the detector at its location.

B. Physical Attack Results

We conducted a physical-domain adversarial attack on both indoor simulation platforms and real-world street scene. The visualization results of the indoor simulation platform are shown in Fig. 10. Results show that our adversarial patches maintain attack efficacy in the complicated physical world,

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successfully transferring from the digital domain to the physical domain.

As illustrated in Fig. 11, visualization results in real-world scenarios demonstrate that the adversarial patch effectively deceives an aerial object detector. Furthermore, our patch exhibits cross-frame generalization, maintaining its efficacy in consecutive frame attacks and broadening its practical applicability. However, compared to the superior stealthiness in the digital domain shown in Fig. 4, the printed patches suffer from degraded stealthiness performance, because of the color discrepancies between digital and physical patches due to printer limitations, dynamic environmental color variations caused by fluctuating illumination, and weather conditions.

C. Robustness of Adversarial Patches in Physical World

The physical robustness is evaluated through multiple conditions, with results shown in Tables IX and X. At flight altitude of 30 and 35 m, the ASR exceeds 90%, confirming significant attack efficacy and demonstrating the robustness of our adversarial patches across varying altitudes. However, when the shooting altitude increases to 40 m, ASR rapidly drops. When the camera is far away from the patches, the number of pixels occupied by the patches in the image decreases, which reduces the adversarial perturbations on the image and leads to a decrease in the patch attack performance. Notably, at a lower camera shooting height of 25 m, the ASR decreases from over 90% to around 50%, mainly due to texture distortion caused by the physical printing process.

The results of attack performance across different camera perspectives show that adversarial patches exhibit strong robustness within the range of camera shooting angles of $60^\circ \sim 90^\circ$. Robustness of cross-perspectives stems from two key design elements in the training phase: 1) deformation modeling via EoT to simulate geometric distortions under varying shooting perspectives and 2) full-image patch rendering strategy that forces patches to learn texture features of target objects in off-center spatial positions under oblique viewing conditions.

Experiments under different environmental lighting conditions and motion stats have also been conducted, and results in Table X show that our physical printed adversarial patches still exhibit robustness under these different conditions.

VI. Conclusion and Future Work

We investigate the vulnerability of aerial object detection algorithm and propose a framework for generating DSAPs. Attackers can embed these patches into scenes to create a nonexistent object on an aerial object detector. An aerial scene-based source-domain image set is constructed to establish environmental priors, and a generator is utilized for generating adversarial patches, while a discriminator is designed to compare the discrepancy between generated patches and source-domain images, thereby ensuring that the generated patches maintain visual consistency with the scene background. Notably, the trained generator demonstrates the capability to directly map distinct latent vectors to diverse adversarial patches, overcoming the limitation of conventional methods that typically produce only one single patch per training cycle. Extensive digital-domain experiments demonstrate that our adversarial patches exhibit superior stealthiness compared to conventional approaches, while maintaining robustness against mainstream defense mechanisms such as adversarial fine-tuning and patch segmentation. Physical-domain validations further confirm successful transferability of the patches from the digital domain to the real world, which substantiates the actual security threats posed by our method to practical aerial detection systems. Our future research will focus on: 1) adversarial patch rendering techniques based on aerial imagery that are more in line with the real world and 2) universal adversarial patch against different detection models. We hope this work will contribute to advancing the robustness of aerial perception algorithms.

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